

Introduction to Quantitative Biology

Lecture Notes: Towards Gene Expression Dynamics

From Cell Biology by the Numbers

September 2026

1. Kleiber's law: how does metabolic rate scale with body size?

We begin by defining the quantities used in the scaling argument:

- M : body mass of the organism;
- N_{cell} : total number of cells in the organism;
- B : whole-organism metabolic rate, or metabolic power;
- P : metabolic rate per cell;
- α : allometric scaling exponent, with $\alpha \approx 3/4$.

If the characteristic mass per cell is approximately constant across animals, then total body mass scales with cell number:

$$M \propto N_{\text{cell}}.$$

Kleiber's law states that whole-organism metabolic rate scales with body mass approximately as

$$B \propto M^\alpha, \quad \alpha \approx \frac{3}{4}.$$

The metabolic rate available per cell is therefore

$$P \equiv \frac{B}{N_{\text{cell}}}.$$

Because $N_{\text{cell}} \propto M$,

$$P = \frac{B}{N_{\text{cell}}} \propto \frac{B}{M} \propto \frac{M^\alpha}{M} = M^{\alpha-1}.$$

Using $\alpha = 3/4$,

$$\boxed{P \propto M^{-1/4}}.$$

Thus, although the **total** metabolic rate B increases with animal size, the metabolic rate **per cell** decreases with body mass.

Comment: Larger animals have a lower metabolic rate per cell, or equivalently a lower metabolic rate per unit mass.

References: West, B. J. *Systems* **2**, 89–118 (2014), Fig. 1; data from Schmidt-Nielsen (1984), CC BY 3.0. West, Brown & Enquist, *Science* **276**, 122–126 (1997).

2. If a human cell needs about 10 days to make enough protein for one division, how long would a mouse cell need?

In the previous lecture, we estimated that if a human cell devoted all of its available energy to protein production, making enough protein for one cell division would take roughly

$$T_{\text{dbl}}^{\text{human}} \sim 10^6 \text{ s} \sim 10 \text{ days}.$$

Question

If we apply the metabolic scaling above to a mouse cell, how long should the corresponding timescale be?

- A. A few hours
- B. A few days
- C. A few weeks
- D. A few months

From Kleiber's law,

$$P \propto M^{-1/4}.$$

If the energy requirement for producing the material needed for one division is comparable, then the required time is inversely proportional to the available metabolic power per cell:

$$T_{\text{dbl}} \propto \frac{1}{P} \propto M^{1/4}.$$

Therefore,

$$\frac{T_{\text{dbl}}^{\text{mouse}}}{T_{\text{dbl}}^{\text{human}}} = \left(\frac{M_{\text{mouse}}}{M_{\text{human}}} \right)^{1/4}.$$

Using

$$M_{\text{mouse}} \sim 25 \text{ g}, \quad M_{\text{human}} \sim 100 \text{ kg} = 10^5 \text{ g},$$

we obtain

$$\frac{T_{\text{dbl}}^{\text{mouse}}}{T_{\text{dbl}}^{\text{human}}} \sim \left(\frac{25}{10^5} \right)^{1/4} \approx 0.13 \sim \frac{1}{10}.$$

Hence,

$$T_{\text{dbl}}^{\text{mouse}} \sim 10 \text{ days} \times \frac{1}{10} \sim 1 \text{ day}.$$

So the order-of-magnitude answer is

B. A few days.

3. How do the human and *E. coli* genomes compare?

3.1 Genome size versus gene number

Question

How much larger is the human genome than the *E. coli* genome, and how much larger is the human protein-coding gene count?

- A. 5000:1
- B. 500:1
- C. 50:1
- D. 5:1

Use the rounded lecture values

$$L_{\text{human}} \sim 3 \text{ Gbp}, \quad L_{\text{E.coli}} \sim 5 - 6 \text{ Mbp}.$$

Since

$$3 \text{ Gbp} = 3000 \text{ Mbp},$$

we obtain

$$\frac{L_{\text{human}}}{L_{\text{E.coli}}} \sim \frac{3000}{5 - 6} \sim 5 \times 10^2.$$

Thus the genome-size ratio is approximately

500 : 1,

so the answer is **B**.

Now compare protein-coding gene numbers:

$$N_{\text{genes, human}} \sim 2 \times 10^4, \quad N_{\text{genes, E.coli}} \sim 4 \times 10^3.$$

Therefore,

$$\frac{N_{\text{genes, human}}}{N_{\text{genes, E.coli}}} \sim \frac{2 \times 10^4}{4 \times 10^3} \sim 5.$$

Thus the gene-number ratio is approximately

$$\boxed{5 : 1},$$

so the answer is **D**.

The human genome is therefore about $500\times$ larger, whereas the number of protein-coding genes is only about $5\times$ larger. The mismatch between these ratios is itself about

$$\frac{500}{5} \sim 100.$$

3.2 Where does this roughly 100-fold difference come from?

A compact bacterial genome is dominated by protein-coding sequence. Human DNA contains much more sequence that does not directly encode proteins. Two major contributors emphasized here are:

- **introns:** a rough lecture-scale estimate is intron : exon $\sim 30 : 1$
- **transposable elements:** approximately 50% of the human genome is derived from transposable elements.

Regulatory and other non-coding sequences add further genomic DNA. Therefore, increasing genome size does not require a proportional increase in protein-coding gene number.

Reference: BioNumbers BNID 110283 for the approximate composition of the human genome; BNID 105750 for the non-coding fraction of the *E. coli* K-12 genome.

3.3 Genome size versus number of genes

Organism	Genome size (kbp)	Protein-coding genes
HIV	10	9
Influenza	14	11
<i>Buchnera</i>	640	610
<i>S. aureus</i>	2,900	2,800
<i>V. cholerae</i>	4,000	3,900
<i>E. coli</i>	4,600	4,200
<i>S. cerevisiae</i>	12,000	5,600
<i>C. elegans</i>	100,000	20,000
<i>M. musculus</i>	3,000,000	20,000

The gene counts in this section explicitly refer to **protein-coding genes**, excluding RNA-coding regions.

Comment: For the organisms listed before *S. cerevisiae*, genome size divided by gene number is consistently of order

$$\frac{L_{\text{genome}}}{N_{\text{genes}}} \sim 1 \text{ kb/gene.}$$

This can be checked directly from the table. For example,

$$\frac{4600 \text{ kb}}{4200 \text{ genes}} \approx 1.1 \text{ kb/gene}$$

for *E. coli*. This agrees with a simple coding estimate: a typical protein is about 300 amino acids long, and each amino acid requires 3 nucleotides,

$$300 \text{ aa} \times 3 \frac{\text{nt}}{\text{aa}} \approx 900 \text{ nt} \sim 1 \text{ kb.}$$

4. How long does it take to diffuse across a cell?

For ordinary diffusion in d dimensions,

$$\langle r^2 \rangle = 2dD\tau,$$

where

- D is the diffusion coefficient;
- τ is the elapsed time;
- $\sqrt{\langle r^2 \rangle}$ is the root-mean-square displacement.

Therefore,

$$1\text{D: } \langle x^2 \rangle = 2D\tau,$$

$$2\text{D: } \langle r^2 \rangle = 4D\tau,$$

$$3\text{D: } \langle r^2 \rangle = 6D\tau.$$

For three-dimensional diffusion,

$$R \equiv \sqrt{\langle r^2 \rangle} = \sqrt{6D\tau},$$

so

$$\boxed{\tau = \frac{R^2}{6D}}.$$

The numerical prefactor depends on dimensionality, but the central scaling result is always

$$\boxed{\tau \propto R^2}.$$

Useful intracellular diffusion coefficients are

$$D_{\text{protein}} \sim 10 \mu\text{m}^2/\text{s},$$

and

$$D_{\text{glucose}} \sim 100 \mu\text{m}^2/\text{s}.$$

4.1 How long does it take to diffuse across a bacterial cell?

- A. μs
- B. ms
- C. s
- D. min

For a protein diffusing across a characteristic bacterial distance,

$$R \sim 1 \mu\text{m}, \quad D_{\text{protein}} \sim 10 \mu\text{m}^2/\text{s}.$$

Then

$$\tau_{\text{bacteria}} \sim \frac{(1 \mu\text{m})^2}{6(10 \mu\text{m}^2/\text{s})} = \frac{1}{60} \text{ s} \approx 1.7 \times 10^{-2} \text{ s}.$$

Therefore,

$$\boxed{\tau_{\text{bacteria}} \sim 10 \text{ ms}},$$

so the answer is **B. ms**.

Reference: BioNumbers BNID 103801 gives a protein diffusion time across an *E. coli* cell of roughly 0.01 s.

4.2 How long does it take to diffuse across a human cell?

- A. μs
- B. ms
- C. s
- D. min

Take a characteristic mammalian-cell distance

$$R \sim 20 \mu\text{m}$$

and use

$$D_{\text{protein}} \sim 10 \mu\text{m}^2/\text{s}.$$

Then

$$\tau_{\text{human cell}} \sim \frac{(20 \mu\text{m})^2}{6(10 \mu\text{m}^2/\text{s})} = \frac{400}{60} \text{ s} \sim 10 \text{ s}.$$

Therefore,

$$\boxed{\tau_{\text{human cell}} \sim 10 \text{ s}},$$

so the answer is **C. s.**

Comment: These calculations estimate the time required to diffuse a characteristic **distance**. R is a root-mean-square displacement, not the time required to arrive at one particular target or location. The diffusion coefficients used here are effective values under intracellular, physiological conditions.

5. How long does it take to diffuse along a 10 cm axon?

Question

How long does it take a protein-sized molecule to diffuse a distance of 10 cm?

- A. 1 min
- B. 1 h
- C. 1 day
- D. 1 month
- E. A few years

Convert the distance:

$$10 \text{ cm} = 10^5 \mu\text{m}.$$

Using the same characteristic protein diffusion coefficient,

$$D_{\text{protein}} \sim 10 \mu\text{m}^2/\text{s},$$

we obtain

$$\tau \sim \frac{(10^5 \mu\text{m})^2}{6(10 \mu\text{m}^2/\text{s})} = \frac{10^{10}}{60} \text{ s} \approx 1.7 \times 10^8 \text{ s}.$$

Since

$$1 \text{ year} \approx 3 \times 10^7 \text{ s},$$

this gives

$$\boxed{\tau \sim 5 \text{ years}}.$$

So the answer is **E. A few years.**

This enormous timescale follows directly from

$$\tau \propto R^2.$$

Increasing the diffusion distance by a factor of 10 increases the diffusion time by a factor of $10^2 = 100$.

Passive diffusion is therefore adequate over micrometre-scale cellular distances but not for long-distance transport along an axon. Cells instead use directed transport along the cytoskeleton by molecular motors.

For comparison, a molecular motor moving at roughly

$$v \sim 1 \text{ } \mu\text{m/s}$$

would traverse 10 cm in

$$t \sim \frac{10^5 \text{ } \mu\text{m}}{1 \text{ } \mu\text{m/s}} = 10^5 \text{ s} \sim 1 \text{ day}.$$